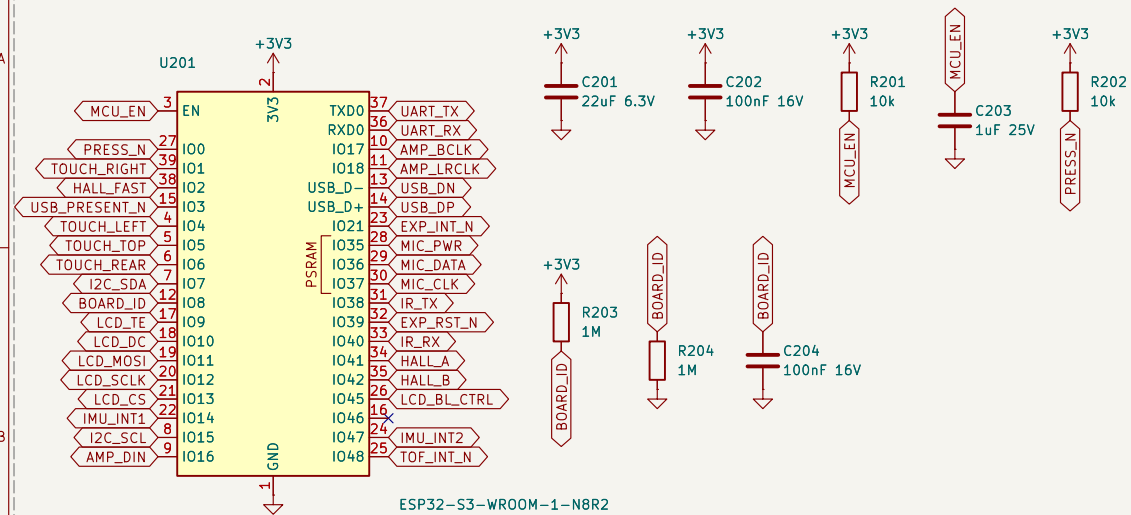






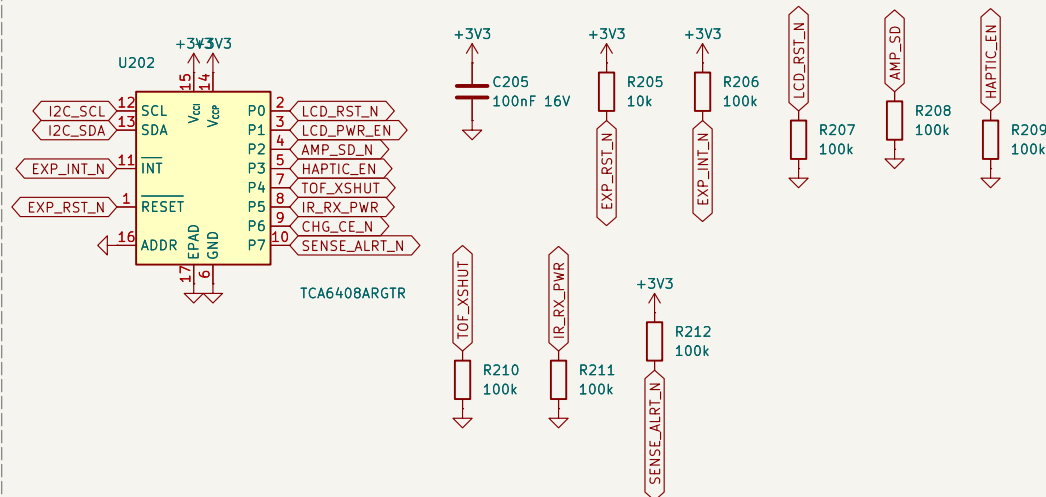
## COMPUTE

## MCU



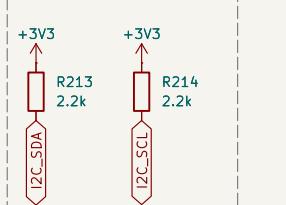
ESP32-S3-WROOM-1-N8R2 on B.Cu at 6 o'clock, antenna over the board-edge notch (no copper beneath). 22 uF + 100 nF at 3V3. EN: 10k/1 uF RC. GPIO0 = face press (BOOT). Unused pins NC or test pads.

## EXPANDER



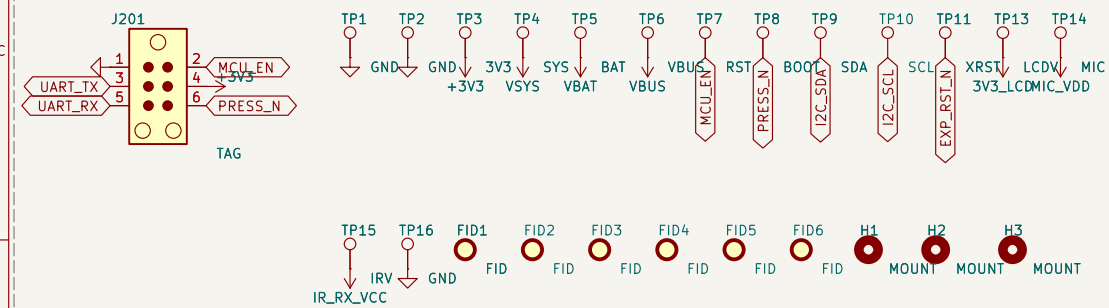
TC6408A at 0x20. Outputs default off through 100k pull-downs; inputs pulled up. Firmware writes the output register before switching pins to outputs (register resets to 0xFF).

## 12C



One 400 kHz bus. 2.2k pull-ups (ODD JOBS 30): ~6 devices, short traces.

## SERVICE



Tag-Connect TC2030-NL (no part fitted): 1 GND, 2 EN, 3 TXD0, 4 3V3, 5 RXD0, 6 GPIO0 – recovery without USB (ODD JOBS 33–35, 157). Pinout chosen so TXD0/RXD0 run straight from the module pins. Probe pads: rails beside the charger, comms in one field, switched rails at their sources (36–39).

ESP32-S3 module, GPIO expander, boot/reset, service and test access

## ODD JOBS

Sheet: /Compute/  
File: compute.kicad\_sch

Title: MAO\_MAIN A0 - Compute

Size: A3

Date: 2026-10-04

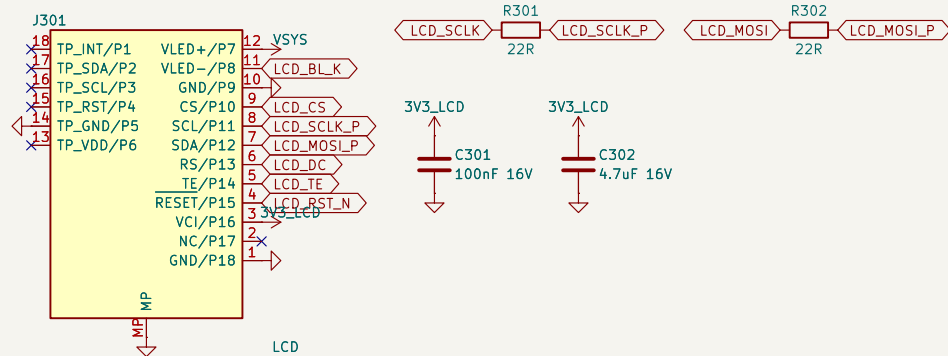
Rev: A0

KiCad E.D.A. 10.0.6

Id: 3/6

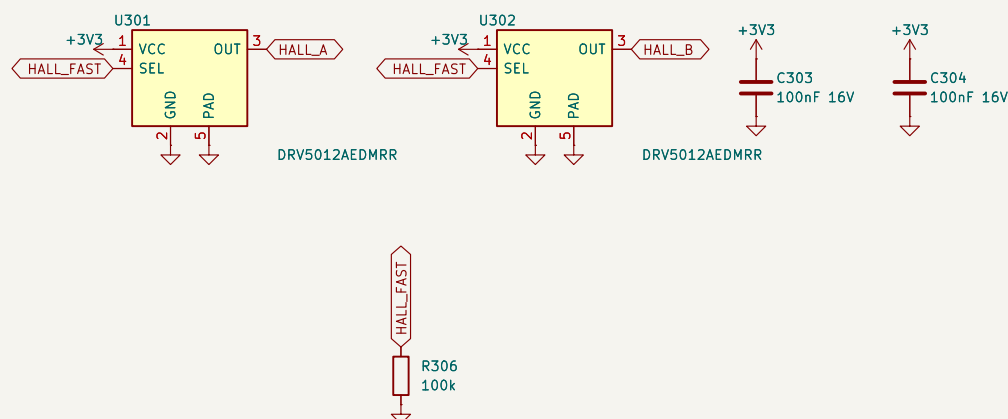
## INTERFACE

## DISPLAY



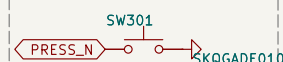
Plug-in display: 1.28" round GC9A01 panel (Winstar WF0128B7YAA4DNN0, 240x240 IPS) on an 18-pin 0.5 mm FPC tail, folded under the panel into a 1.0 mm high back-flip connector with top and bottom contacts, so any panel of this 18-pin standard plugs in either way and can be swapped without solder. Pins 1-6 serve the touch controller of touch variants: not wired (MAO's face is a pressed window), TP\_GND to GND, TE to GPIO9 for tear-free frames, 22R on SCLK/MOSI damps 80 MHz edges (ODD JOBS 29). Panel logic on the switched 3V3\_LCD rail, backlight anode on VSYS through the constant-current driver; RST held low until firmware. The stock 70 mm tail S-folds in the face carrier, drops through the board slot at 9 o'clock and plugs into J301 on B.Cu.

## RING DIAL



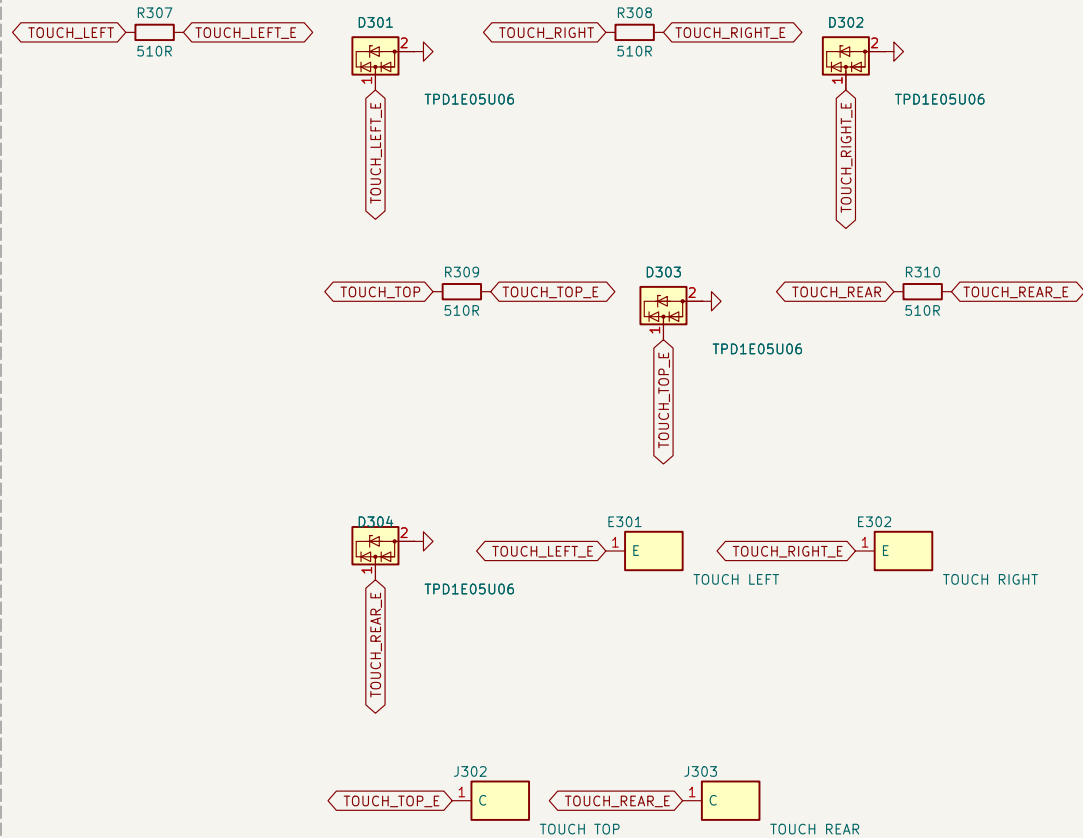
Two DRV5012 Hall latches under the ring's 30-pole ferrite strip, 6 deg apart (half a pole = 90 deg electrical); A/B quadrature identical to the LCDkit EC11 (2 transitions per detent, 30 detents). HALL\_FAST low = 20 Hz sampling, 1.6 uA (sleep); high = 2.5 kHz (awake).

FACE PRESS



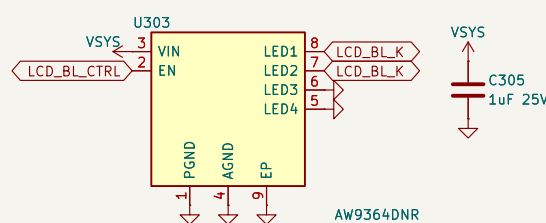
ALPS SKQG on F.Cu at the centre, under  
the panel: the floating face presses it.  
GPI00 = BOOT strap: no capacitor here.

## BODY TOUCH



Four ESP32-S3 touch channels. 510R within 1 mm of the module (Espressif guide); 0.42 pF ESD at the electrode. LEFT/RIGHT: copper arcs at the rim; TOP/REAR: gold springs to printed or foil electrodes.

**BACKLIGHT**



Panel LEDs (one group of two) need VLED+ 3.0/3.2/3.4 V at 40 mA (Winster spec 4.2); more than 3V3 can drive. AW9364 constant-current sinks from V3S4 (+40~50 mV dropout): LED1+LED2 on VLED- = 2 x 20 mA = 40 mA nominal (33~47 mA ~ -17.5 % part tolerance), set by the part, not by firmware. 16 steps by EN pulses on GPIO405; its 150k EN pull-down keeps the panel dark from reset (ODD J0B5 116). Full 40 mA down to V3S5 3.4 V on the typical 3.2 V bin, 3.6 V on the 3.4 V bin (sim S3); dimmer below.

Round display, backlight, ring dial, face press, body touch

## ODD JOBS

Sheet: /Interface/

File: interface.kicad\_sch

**Title: MAO\_MAIN A0 – Interface**

Size: A3

Date: 2026-10-04

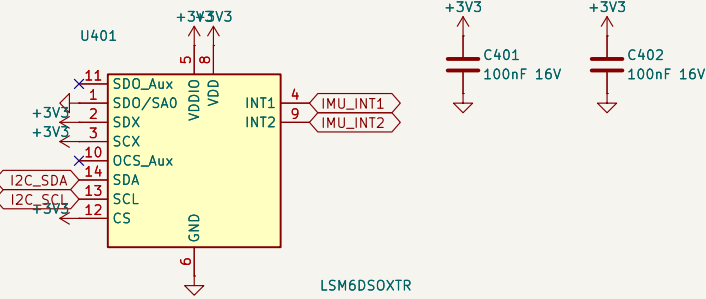
Rev: A0

KiCad E.D.A. 10.0.6

Id: 4/6

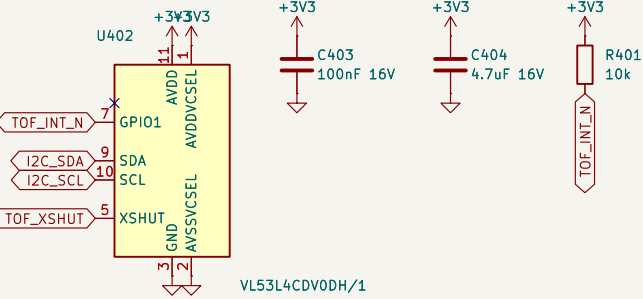
SENSE

IMU



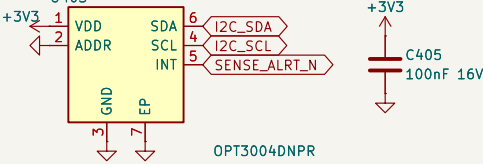
LSM6DSOX at 0x6A (SA0 low), I2C (CS high), SDx/SCx to VDDIO keeps the land compatible with the LSM6DS3TR-C fallback. INT1/INT2 push-pull, configured active-low for ext1 wake.

PROXIMITY



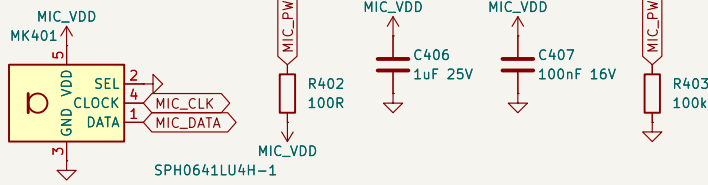
VL53L4CD time-of-flight, 0-1.3 m, under the window border. XSHUT from the expander (off by default); GPIO1 open-drain with the 10k pull-up ST recommends. AVDD decoupled 4.7 uF + 100 nF per ST.

AMBIENT LIGHT



OPT3004 at 0x44, human-eye response, under the window border. INT joins SENSE\_ALERT\_N.

MICROPHONE



SPH0641 bottom-port PDM mic on B.Cu, port through the board (0.5 mm hole) to the top. Powered from GPIO35 through 100R/1 uF only while listening: it draws 80 uA even with the clock stopped. SEL low.

RAIL SOURCE DECLARATIONS

IMU, proximity, ambient light, microphone

ODD JOBS

Sheet: /Sense/  
File: sense.kicad\_sch

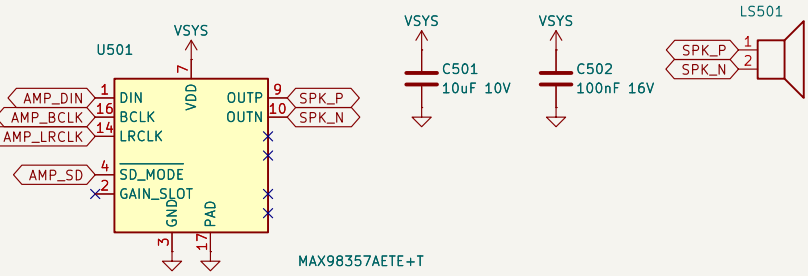
Title: MAO\_MAIN A0 - Sense

Size: A3 Date: 2026-01-06  
KiCad E.D.A. 10.0.6

Rev: A0  
Id: 5/6

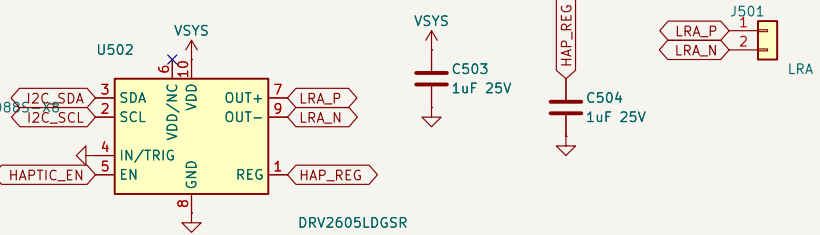
FEEDBACK

SPEAKER AMP



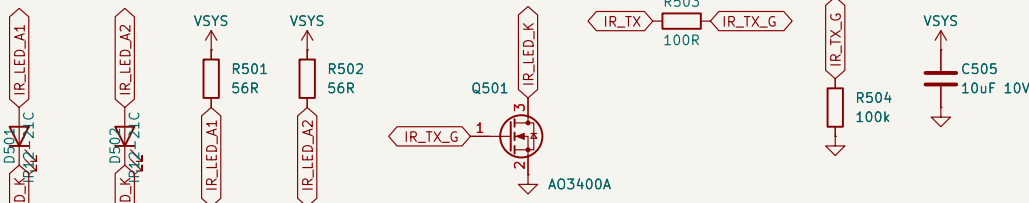
MAX98357A 12S class-D from VSYS. SD\_MODE from the expander through 2.2k (datasheet: series R when VDDIO can exceed VDD): low = 0.6 uA shutdown, 3.2 V = left channel. GAIN\_SLOT open = 9 dB (datasheet state, deliberate). Speaker: Same Sky CMS-150803-0885-X8 (15 x 8 x 3 mm, 8 ohm, 0.8 W) in the base at 9 o'clock: its own spring contacts press on two pads on B.Cu (LS501).

HAPTIC



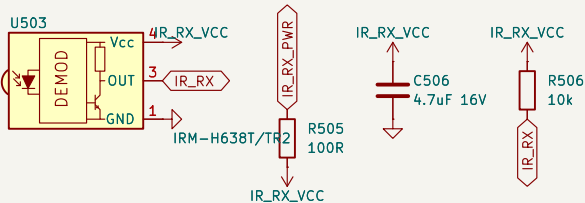
DRV2605L closed-loop LRA driver from VSYS. I2C 0x5A. EN from the expander (4 uA off). IN/TRIG low: playback by I2C. Pin 6 VDD/NC left open (SLOS854D: optional supply, tie to VDD or float). LRA LD0832AA (235 Hz, 1.8 Vrms) glued to the base, leads to J501.

IR TRANSMIT



Two side-emitting 940 nm LEDs at the back edge (B.Cu), each with its own 56R from VSYS (26-59 mA pulses at 38 kHz, 33 % duty: under the 65 mA rating at any VSYS up to 4.5 V and any Vf). A03400A low side, gate pulled down and driven from GPIO38, a pin without a reset pull-up: dark at reset (117).

IR RECEIVE



IRM-H638T (as LCDkit), top-view, looking up through the window border. Powered from expander P5 through 100R/4.7 uF only when listening (0.4 mA). OUT pull-up goes to the switched supply, never to +3V3, so an unpowered receiver is never back-powered and GPIO40 reads a defined low while it is off.

SPEAKER AMP



MAX98357A 12S class-D from VSYS. SD\_MODE from the expander through 2.2k (datasheet: series R when VDDIO can exceed VDD): low = 0.6 uA shutdown, 3.2 V = left channel. GAIN\_SLOT open = 9 dB (datasheet state, deliberate). Speaker: Same Sky CMS-150803-0885-X8 (15 x 8 x 3 mm, 8 ohm, 0.8 W) in the base at 9 o'clock: its own spring contacts press on two pads on B.Cu (LS501).

RAIL SOURCE DECLARATIONS  
Speaker amplifier, haptic driver, IR transmit and receive

ODD JOBS

Sheet: /Feedback/  
File: feedback.kicad\_sch

Title: MAO\_MAIN A0 - Feedback

Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-12-18  
Rev: A0

Id: 6/6